

Date : 04/10/2016
 Time : 1 hr. 30 min

Max.Marks:80

General Instructions to the candidate

Please write your answers legibly and neatly in the answer booklet provided only.
 Symbols, constants, terminology used have their usual meaning unless specified specially.
 There are a total of **FOUR** questions and all are compulsory. Each carries marks as indicated.
 Answer the **subparts** of a question at one place and in the order in which they appear.
 Preferably start each question on a **fresh page** however sub parts can be written on same side.

Common data: Use the following common data if not mentioned specifically in the question

For 0.25 μ m Technology node $V_{DD} = 3.3V$, $V_{Thermal} = 25.9mV$

- Use long channel approximation and lambda based design rules unless specially mentioned.
- Use square law current equation unless specifically mentioned.
- Neglect body effect, channel length modulation if not mentioned specifically.
- The body of all PMOS is tied to V_{DD} while all NMOS is tied to ground unless specially mentioned.

$V_{DD}=3.3V$, $L_{min}=250nm$, $\mu_n C_{ox}=120 \mu A/V^2$, $\mu_p C_{ox}= 40 \mu A/V^2$, $V_{TON}=0.7V$, $V_{TOP}= -0.9V$

Q1 A static CMOS inverter is designed such that $W_p=30 \mu m$, $W_n=12 \mu m$, $L_p=L_n=0.25 \mu m$.

- i. Assume an ideal square wave as input, neglect parasitic capacitances and estimate the worst case delay using the average current method if the load capacitance is 50 fF.
- ii. What should be the width of a PMOS (assume $L_p=0.25 \mu m$) transistor that should be added in parallel to the pull up network of the above inverter in order to make its VTC symmetric.
- iii. Now, Determine pullup to pull down ration of this inverter. Will its logical effort 'g' be different from that of a CMOS inverter which is designed in the ratio of mobility (μ_n/μ_p) only? justify
- iv. What is the minimum allowable value of V_{DD} which still retains the inverter action?

[20]

Q2 (A) Implement 2 Input XOR gate using pseudo nMOS logic with minimum number of transistors. The inputs are available only in true form.

(B) Fig. 1 shows a complex Or And Invert gate. The (W/L) value of all the pull down transistors are 5/1. Find out the following:

- i. Logic function F.
- ii. the equivalent W/L of the pull down network
- iii. Voltage V_{OH}
- iv. Worst case value of V_{OL} if $(W/L)_{load}$ is 2/1 . (ignore body bias and channel length modulation effect)
- v. What will be impact of increasing the $(W/L)_{load}$ on V_{OL} and region of operation of load transistor?

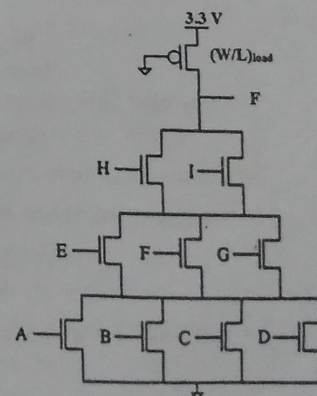


Fig.1

[20]

PTO

Q3 The data path from adder block to the memory within a microprocessor involves 500 ps of gate delay and 500 ps of wire delay along this critical path. The chip is scaled using constant field scaling with scaling factor $S=2$. For interconnects, only the height of the wires is scaled down with a scaling factor $S=2$.

- Calculate the gate and wire delays of the scaled data path.
- Estimate, whether there is an overall delay improvement, Justify?
- Also calculate, increment/decrement in the dynamic power consumption, if there are 50 K transistors on the chip with an activity factor 25% at any given time. (Ignore wire and parasitic capacitances)

[20]

Q4 The layout shown in figure 1 all PMOS transistors have width $W_p=10\text{ }\mu\text{m}$, and all NMOS transistor has width $W_n=5\text{ }\mu\text{m}$. The length for both NMOS and PMOS transistor are equal (i.e. $L_n=L_p=1\text{ }\mu\text{m}=2\lambda$).

- Draw the equivalent transistor level circuit indicating the W/L ratios and identify the logic of operation.
- If the length of diffusion for each transistor $L_{\text{Diff}}=20\lambda$ and $C_{j0,sw,n}=C_{j0,sw,p}=10^{-10}\text{ F/m}$, $C_{j0,n}=C_{j0,p}=10^{-4}\text{ F/m}^2$, $C_{ox}=5\times 10^{-3}\text{ F/m}^2$. Identify the type of capacitance and also calculate the capacitances at Output node 'Y'. (Ignore the metal wire and overlap capacitances, ignore perimeter capacitance facing the poly gate)

(20)

